$$\frac{\omega^2}{c^2} n^2(\omega) = k^2 \quad , \quad n^2(\omega) = (1+\chi_e)(1+\overset{\to 0}{\chi_m}) \Rightarrow n(\omega) \equiv n_\alpha(\omega)$$

depends on polarization.

Linearly polarized wave

$$\vec{E}_1(z,t) = E_0 \cos(kz-\omega t)\hat{x} =$$

$$= \frac{1}{2}\left(\vec{E}_{\alpha=+1} + \vec{E}_{\alpha=-1}\right)$$

$$\vec{E}_\alpha = E_0\left(\cos(kz-\omega t)\hat{x} + \underbrace{\cos\left(kz-\omega t+\frac{\alpha\pi}{2}\right)\hat{y}}_{-\alpha \sin(kz-\omega t)\hat{y}}\right)$$

$$\vec{E}_\alpha(z,t) = Re\left\{(\hat{x}+i\hat{y})\, e^{i(kz-\omega t)\alpha}\right\} = Re\left\{(\hat{x}+i\hat{y})\left(\cos(kz-\omega t)+i\alpha\sin(kz-\omega t)\right)\right\}$$

$$= \left(\cos(kz-\omega t)\,\hat{x} - \alpha\sin(kz-\omega t)\hat{y}\right)E_0 .$$

For the case of $n''=0$, $k = \frac{\omega n(\omega)}{c} \in$ Reals

$$\Rightarrow \quad k = k_\alpha = \frac{\omega}{c} n_\alpha$$

$\Rightarrow$ let at $z=0$ $\vec{E}_1(0) = E_0 \cos(\omega t)\hat{x}$ $\quad (\alpha=+1)$

then at z $\quad \vec{E}_1(z,t) = \frac{1}{2}E_0\, Re\left\{(\hat{x}+i\hat{y})\, e^{i(k_+ z-\omega t)} + \right.$

$$\left. + (\hat{x}+i\hat{y})\, e^{-i(k_- z-\omega t)}\right\} = \frac{1}{2}E_0\, Re\left\{(\hat{x}+i\hat{y})\cdot\right.$$

$$\left.\cdot\left[\cos(k_+ z-\omega t) + \cos(k_- z-\omega t) + \right.\right.$$

$$\left.\left. + i\sin(k_+ z-\omega t) - i\sin(k_- z-\omega t)\right]\right\} =$$

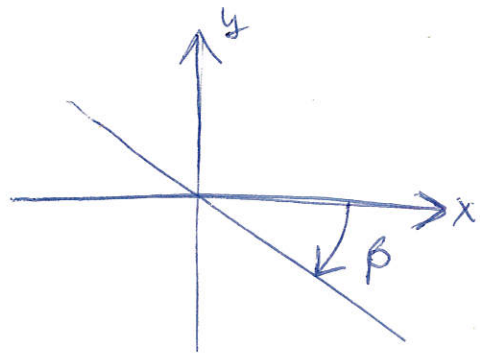
$$= \frac{1}{2} E_0 \Big\{ \big(\cos(k_+ z - \omega t) + \cos(k_- z - \omega t)\big)\hat{x}$$

$$- \big(\sin(k_+ z - \omega t) - \sin(k_- z - \omega t)\big)\hat{y} \Big\} =$$

$$= \frac{1}{2} E_0 \Big\{ 2\cos\left(\frac{(k_+ + k_-)z}{2} - \omega t\right) \cos\frac{(k_+ - k_-)z}{2}\ \hat{x}$$

$$- 2 \sin\frac{(k_+ - k_-)z}{2} \cos\left(\frac{(k_+ + k_-)z}{2} - \omega t\right)\hat{y} \Big\}$$

$$= E_0 \cos\left(\frac{(k_+ + k_-)}{2} z - \omega t\right) \left(\cos\frac{(k_+ - k_-)z}{2}\ \hat{x} - \sin\frac{(k_+ - k_-)z}{2}\ \hat{y}\right)$$



$\vec{E}_1$ has rotated by

$$\beta = \frac{(k_+ - k_-)z}{2}$$

far from resonance frequencies

$n \approx 1 + \frac{1}{2}\chi$

$$\frac{\beta}{z} = \frac{\omega}{2c}(n_+ - n_-) = \frac{\omega}{2c}\left(1 + \frac{\chi_+}{2} - \left(1 + \frac{\chi_-}{2}\right)\right) =$$

$$= \frac{\omega}{4c}\left(\frac{\varepsilon_0 \omega_p^2}{\omega_0^2 - \omega^2 - \Omega_c \omega} - \frac{\varepsilon_0 \omega_p^2}{\omega_0^2 - \omega^2 + \Omega_c \omega}\right) =$$

$$= \frac{\omega}{4c} \varepsilon_0 \omega_p^2 \frac{2\Omega_c \omega}{(\omega_0^2 - \omega^2)^2 - \Omega_c^2 \omega^2} \sim \frac{\omega^2 \omega_p^2}{(\omega_0^2 - \omega^2)^2} \cdot B_0$$

Plasma freq. $\omega_p^2 = \frac{Nq^2}{\varepsilon_0 m}$

Polarization axes has rotated by $\beta \sim B_0 z$.

~~Before~~

Derivation of polarization-dependent susceptibility
(similar to the calculations done in class)

P1

Equation of motion

$$\ddot{\vec{r}} = -\omega_0^2 \vec{r} + \frac{q}{m} \dot{\vec{r}} \times \vec{B}_0 + \frac{q}{m} \vec{E}_\alpha(z,t)$$

$$\dot{\vec{r}} \times \vec{B}_0 = \hat{x}\, \dot{y} B_0 - \hat{y}\, \dot{x} B_0$$

Since wavelength $\lambda \gg$ typical r, we can neglect z-dependence of $\vec{E}_\alpha \to E_0 \left[\cos(\omega t)\hat{x} + \cos(-\omega t + \alpha\frac{\pi}{2})\hat{y}\right]$

$\cos(\omega t - \alpha \frac{\pi}{2}) = \sin(\omega t) \sin(\alpha \frac{\pi}{2}) = \alpha \sin(\omega t)$.

for $\alpha = \pm 1$.

So $\vec{E}_\alpha(t) = E_0 \left[\cos(\omega t)\hat{x} + \alpha \sin(\omega t)\hat{y}\right]$.

(1) $\ddot{x} - \frac{qB_0}{m}\dot{y} + \omega_0^2 x = \frac{q}{m} E_0 \cos(\omega t)$

(2) $\ddot{y} + \frac{qB_0}{m}\dot{x} + \omega_0^2 y = \frac{q}{m} E_0 \alpha \sin(\omega t)$

Consider circular variables: $\begin{cases} \xi = x + iy \\ \eta = x - iy \end{cases}$

Multiply (2) by (i) and add to (1):

$$\ddot{\xi} - \frac{qB_0}{m}\underbrace{(\dot{y} - i\dot{x})}_{(-i)(\dot{x}+i\dot{y})} + \omega_0^2 \xi = \frac{qE_0}{m}\underbrace{\left(\cos(\omega t) + i\alpha \sin(\omega t)\right)}_{e^{i\alpha\omega t} \text{ for } \alpha = \pm 1}$$

$$\Rightarrow \ddot{\xi} + i\frac{qB_0}{m}\dot{\xi} + \omega_0^2 \xi = \frac{qE_0}{m} e^{i\alpha\omega t}$$

Look for solution $\xi(t) = \xi_0 e^{i\omega_\alpha t}$

$$\Rightarrow \dot{\xi} = i\omega_\alpha \xi, \quad \ddot{\xi} = -\omega_\alpha^2 \xi$$

$$\left[-\omega_\alpha^2 + i\frac{qB_0}{m}(i\omega_\alpha) + \omega_0^2\right]\xi_0 e^{i\omega_\alpha t} = \frac{qE_0}{m}e^{i\alpha\omega t}$$

$$\Rightarrow \omega_\alpha = \alpha\omega \quad \text{and}$$

$$\xi_0 = \frac{qE_0/m}{\omega_0^2 - \omega^2 - \alpha\frac{qB_0}{m}\omega}$$

Note that $\frac{qB_0}{m} = \Omega_c$ = cyclotron frequency

Thus for $\underline{\alpha = +1}$:

$$\xi(t) = x(t) + iy(t) = \frac{qE_0/m}{\omega_0^2 - \omega^2 - \Omega_c\omega}e^{i\omega t}$$

$\underline{\alpha = -1}$:

$$x(t) + iy(t) = \frac{qE_0/m}{\omega_0^2 - \omega^2 + \Omega_c\omega}e^{-i\omega t}$$

$\underline{\alpha = +1}$: $$x(t) = \mathrm{Re}(\xi(t)) = \frac{qE_0\cos\omega t}{m(\omega_0^2 - \omega^2 - \Omega_c\omega)}$$

$$y(t) = \mathrm{Im}(\xi(t)) = \frac{qE_0\sin\omega t}{m(\omega_0^2 - \omega^2 - \Omega_c\omega)}$$

$\underline{\alpha = -1}$: $$x(t) = \frac{qE_0\cos\omega t}{m(\omega_0^2 - \omega^2 + \Omega_c\omega)}$$

$$y(t) = \frac{qE_0(-\sin\omega t)}{m(\omega_0^2 - \omega^2 + \Omega_c\omega)}$$

$$\vec{r}(t) = x(t)\,\hat{x} + y(t)\,\hat{y}$$

$$\vec{p}(t) = q\,\vec{r}(t)$$

$\underline{\alpha = +1}$

$$\vec{p}(t) = \frac{q^2 E_0}{m(\omega_0^2 - \omega^2 - \Omega_c \omega)} \left(\cos(\omega t)\,\hat{x} + \sin(\omega t)\,\hat{y}\right)$$

$$= \frac{q^2/m}{\omega_0^2 - \omega^2 - \Omega_c \omega}\, \vec{E}_{\alpha=+1}(t)$$

$$\vec{P} = \underset{\substack{\uparrow \\ \text{concentration of atoms}}}{N}\,\vec{p} = \varepsilon_0 \chi \vec{E}$$

$$\Rightarrow \quad \chi_{\alpha=+1}(\omega) = \frac{N q^2}{m(\omega_0^2 - \omega^2 - \Omega_c \omega)}$$

$\underline{\alpha = -1}$

$$\vec{p}(t) = \frac{q^2 E_0}{m(\omega_0^2 - \omega^2 + \Omega_c \omega)} \left(\cos(\omega t)\,\hat{x} - \sin(\omega t)\,\hat{y}\right)$$

$$= \frac{q^2/m}{\omega_0^2 - \omega^2 + \Omega_c \omega}\, \vec{E}_{\alpha=-1}(t).$$

$$\chi_{\alpha=-1}(\omega) = \frac{N q^2}{m(\omega_0^2 - \omega^2 + \Omega_c \omega)}$$

Re-write denominators: $\omega^2 \pm \Omega_c \omega = \left(\omega \pm \frac{\Omega_c}{2}\right)^2 - \left(\frac{\Omega_c}{2}\right)^2$

But $\frac{1}{2}\Omega_c = \frac{q B_0}{2m} = \Omega_L$ = Larmor frequency.

Both denominators can be written as

$$\omega_0^2 - \omega^2 \mp 2\Omega_L\omega = \omega_0^2 - \left((\omega \pm \Omega_L)^2 - \Omega_L^2\right) =$$

$$= \omega_0^2 + \Omega_L^2 - (\omega \pm \Omega_L)^2 \quad \text{where } (+) \text{ for } \alpha = +1.$$

$$(-) \text{ for } \alpha = -1$$

$$\Rightarrow \quad \chi_\alpha(\omega) = \frac{Nq^2/m}{\omega_0^2 + \Omega_L^2 - (\omega + \alpha\Omega_L)^2}$$

Resonance, when denominator $= 0$, takes place at frequency $(\omega_\alpha + \alpha\Omega_L)^2 = \omega_0^2 + \Omega_L^2$

$$\omega_\alpha = -\alpha\Omega_L + \sqrt{\omega_0^2 + \Omega_L^2}$$

$$\left\{ \begin{array}{ll} \underline{\alpha = +1} & \omega = -\Omega_L + \sqrt{\omega_0^2 + \Omega_L^2} \\ \underline{\alpha = -1} & \omega = \Omega_L + \sqrt{\omega_0^2 + \Omega_L^2} \end{array} \right\} \text{different for different polarizations } \alpha.$$

P.2 <u>Uniaxial dielectric.</u>

$$\varepsilon_{ij} = \varepsilon_0 \begin{pmatrix} \varepsilon_x & 0 & 0 \\ 0 & \varepsilon_x & 0 \\ 0 & 0 & \varepsilon_z \end{pmatrix} \quad , \quad D_i = \sum_j \varepsilon_{ij} E_j \quad \text{but } \vec{B} = \mu_0 \vec{H}.$$

$$\vec{k} \times \vec{E} = \omega \vec{B} = \mu_0 \omega \vec{H}$$

$$\vec{k} \times \vec{H} = - \omega \vec{D}$$

$$\omega \vec{D} = - \vec{k} \times \left(\frac{\vec{k} \times \vec{E}}{\mu_0 \omega} \right) = - \frac{1}{\mu_0 \omega} \left(\vec{k} (\vec{k} \cdot \vec{E}) - k^2 \vec{E} \right)$$

$$\Rightarrow \vec{D} = \frac{\varepsilon_0 c^2}{\omega^2} \left(k^2 \vec{E} - \vec{k} (\vec{k} \cdot \vec{E}) \right)$$

thus
$$\begin{cases} D_i = \dfrac{\varepsilon_0 c^2}{\omega^2} \left(k^2 E_i - k_i (k_j E_j) \right) \\ D_i = \varepsilon_{ij} E_j \end{cases}$$
(sum over repeated indices is assumed here)

$$\Rightarrow \frac{\varepsilon_{ij}}{\varepsilon_0} E_j = \frac{c^2}{\omega^2} \left(k^2 E_i - k_i (k_j E_j) \right) = n^2 E_i - n_i (n_j E_j)$$

where $\vec{k} = \frac{\omega}{c} \vec{n}$, $k_i = \frac{\omega}{c} n_i$

Therefore we obtain matrix equation
$$\left(\frac{\varepsilon_{ij}}{\varepsilon_0} - n^2 \delta_{ij} + n_i n_j \right) E_j = 0$$

Observe that matrix $n_i n_j E_j$ means the following:

$$\begin{pmatrix} n_x^2 & n_x n_y & n_x n_z \\ n_y n_x & n_y^2 & n_y n_z \\ n_z n_x & n_z n_y & n_z^2 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}$$

Therefore
$$\begin{pmatrix} \varepsilon_x - n_z^2 - n_y^2 & n_x n_y & n_x n_z \\ n_y n_x & \varepsilon_x - n_z^2 - n_x^2 & n_y n_z \\ n_z n_x & n_z n_y & \varepsilon_z - n_x^2 - n_y^2 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} = 0$$

Determinant of that matrix <u>must be</u> zero, so that the matrix is <u>not</u> invertible. (Otherwise the above equation simply gives $\vec{E} = 0$ which describes no wave at all.)

(MATHEMATICA is helpful here!)

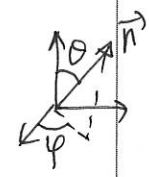
$$\det\left(\downarrow \right) = 0 \text{ gives } (n^2 - \varepsilon_x)\left(\varepsilon_x (n_x^2 + n_y^2) + \varepsilon_z n_z^2 - \varepsilon_x \varepsilon_z\right) = 0.$$

(here $n^2 \equiv n_x^2 + n_y^2 + n_z^2$, of course)

Therefore two solutions:

1) $n^2 = \varepsilon_x \Rightarrow k = \frac{\omega}{c}\sqrt{\varepsilon_x}$, $\omega = \frac{c}{\sqrt{\varepsilon_x}} k$

$V_0 = \frac{c}{\sqrt{\varepsilon_x}}$ is just the ordinary wave.



2) $\varepsilon_x (n_x^2 + n_y^2) + \varepsilon_z n_z^2 = \varepsilon_x \varepsilon_z$

let $\vec{n} = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta) \cdot \tilde{n}(\theta)$

$$\Rightarrow \varepsilon_x \sin^2\theta\, \tilde{n}^2 + \varepsilon_z \cos^2\theta\, \tilde{n}^2 = \varepsilon_x \varepsilon_z$$

$$\frac{\sin^2\theta}{\varepsilon_z} + \frac{\cos^2\theta}{\varepsilon_x} = \frac{1}{\tilde{n}^2} = \frac{1}{\tilde{n}^2(\theta)}$$

or, $$\tilde{n}(\theta) = \sqrt{\frac{\varepsilon_x \varepsilon_z}{\varepsilon_x \sin^2\theta + \varepsilon_z \cos^2\theta}}$$

Thus from $\vec{k} = \frac{\omega}{c}\vec{n} \Rightarrow k = \frac{\omega}{c}|\vec{n}| = \frac{\omega}{c}\tilde{n}$

$$\omega = \frac{c}{\tilde{n}(\theta)} k \Rightarrow V_e(\theta) = \frac{c}{\tilde{n}(\theta)} = c\sqrt{\frac{\cos^2\theta}{\varepsilon_x} + \frac{\sin^2\theta}{\varepsilon_z}}$$

This is extraordinary wave the speed of which depends on angle θ.